

Phạm Khánh Minh Mẫn

Backend Engineer · E-commerce API Integration · LLM-Agent Memory Research

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SUMMARY

Backend engineer with 5+ years in e-commerce, payments and multi-marketplace API integration, mostly in NestJS and TypeScript with event-driven work on AWS. Author of four open-source client libraries for the Shopee, TikTok Shop and Lazada Open APIs, published on npm.

Since 2026, a graduate student in Computer Science (research track) at Danang University of Science and Technology, interested in memory for LLM agents.

EXPERIENCE

DiproTech · Backend Developer 07/2025 - present

Backend work for Japanese clients, often more than one project at a time.

Social App 07/2025 - 03/2026

Newsfeed and content distribution, team of 10.

- Owned the newsfeed backend. The shuffled feed is ordered by a hash of a per-user seed and the post id, so pagination stays stable without storing any order.
- Built the deploy path: dev and staging on Docker Compose over SSH, production on ECS.
- Images are built with a Buildx layer cache and tagged by commit SHA in ECR; a production run registers a new task definition per service and can ship one service without touching the others.

NestJS · TypeScript · PostgreSQL · Redis · AWS ECS / ECR · GitHub Actions · Docker Buildx

Other client projects

- Manufacturing simulation bridged to factory equipment over OPC-UA: added quantity-based batch equipment to the simulator, and carried a schema migration through the five services that share one models package.
- Publishing platform (four apps behind one Cognito sign-in): passkey registration, listing and removal in the shared auth library and the ID service, with API tests.
- Subscription billing on Stripe and PayPal, and an aptitude-test app backend: see Selected systems.

Python · Django · Java · OPC-UA · Nuxt · Next.js · AWS Cognito

Devtify Technologies · Backend Developer 04/2022 - 06/2025

PaymentShield 01/2025 - 06/2025

Auto loan service between lenders, dealers and customers, team of 7.

- Started the codebase: NestJS, Docker, PostgreSQL, layered by responsibility.
- Webhooks publish to SQS and Lambda consumes them, so a slow downstream cannot block the callback.
- Failed jobs retry with backoff and land in a DLQ instead of disappearing.

NestJS · TypeScript · PostgreSQL · AWS Lambda · AWS SQS · BullMQ · Redis

ROUTINE 01/2024 - 03/2025

Orders, inventory and refunds unified across Shopee, Lazada and TikTok Shop, team of 10.

- One order model over three marketplaces, each with its own API shape and failure modes.
- Kept stock and finance in step with the NaviWorld ERP.
- Integrated three payment gateways and three shipping providers.

NestJS · TypeScript · PostgreSQL · Redis · Elasticsearch · BullMQ · Shopee API · TikTok Shop API · Lazada API

BEAUTYBOX / THEFACESHOP / REEBOK 04/2022 - 11/2023

Retail digital transformation for HSVGroup, team of 15.

- Designed the database and REST API.

NestJS · TypeScript · PostgreSQL · Redis · AWS EC2 · Docker

Vietnam People's Army · Military Service 01/2020 - 01/2022

Completed compulsory military service.

D-SOFT JSC · Backend Developer 03/2019 - 12/2019

SunWorld B2B Ticket

Ticket selection, online payment and e-ticket storage, team of 7.

- Designed and built the REST API on PHP / Laravel / SQL Server.

PHP · Laravel · SQL Server

SELECTED SYSTEMS

BattleCatsLab battlecatslab.fun Personal project · public

Data and simulation site for a mobile game - 9,000 generated pages, a deterministic battle engine, and server-authoritative realtime PvP.

- Server owns the sim and ticks it at 30 Hz; clients send deploy intents and render snapshots, so a client cannot fake a result.
- Ranked ELO gate treats same-IP as telemetry only - shared Wi-Fi must not invalidate a real match; the abuse lockout derives from an append-only log, not a mutable counter.
- A golden-vector test pins fixed team pairs to their exact winner and frame count, so an engine change that would desync stored replays fails CI instead of corrupting them silently.

TypeScript · React · Vite · SQLite · Cloudflare Workers · Cloudflare R2 · SSE

Subscription billing on Stripe and PayPal

Recurring payments · DiproTech client

Rails subscription platform for a Japanese publisher: plan upgrades, webhook settlement and tax-compliant receipts, across two payment providers at once.

- Stripe signs the raw body, so the HMAC verifies locally; PayPal needs an OAuth2 token and a call back to verify-webhook-signature - verification itself is a network hop that can fail.
- Both deliver at least once and out of order: a conditional UPDATE decides which delivery settles the payment, and stale events are dropped by timestamp so a late failure notice cannot revive a cancelled subscription.
- The webhook usually beats the browser back, so payment settles there rather than on the success redirect, with the subscription re-read from the gateway API instead of the event payload.
- Dunning differs too: Stripe marks past_due only after its retries are exhausted, while PayPal walks PAYMENT.FAILED -> SUSPENDED -> CANCELLED.

Ruby on Rails 8 · PostgreSQL · Stripe · PayPal Subscriptions · Sidekiq · Redis · RSpec · Heroku

Aptitude-test study app (backend)

EdTech · DiproTech client

FastAPI service on Firestore: question bank, topic taxonomy, per-user progress.

- Rapid mark-difficult / mark-unlearned calls overwrote each other's stats - a write race, not a slow query.
- Fixed by updating counters incrementally instead of recomputing the category, which also dropped a full scan per request.

Python · FastAPI · Firebase Firestore · BigQuery · Cloud Build

OPEN-SOURCE PACKAGES

Shopee API Client (npm: shopee-api-client) - TypeScript client for Shopee Open API v2. Covers seller authorization, token management, orders, products, logistics, and payment escrow.

TikTok Shop API Client (npm: tiktokshops-api-client) - TypeScript API client for TikTok Shop Open API. Covers seller authorization, order APIs, product APIs, and fulfillment APIs.

Lazada API Client (npm: lazada-api-client) - TypeScript API client for Lazada Open API. Covers seller authorization, order APIs, and product APIs.

All-in-One Package (npm: shopee-tiktokshops-lazada-api) - Bundles the Shopee, TikTok Shop and Lazada SDKs into a single dependency, with versions kept in sync automatically.

SKILLS

Programming languages	TypeScript, JavaScript, Python, PHP, Ruby
Backend Frameworks	NestJS, Express, Laravel, Ruby on Rails, Django, FastAPI
Databases & Search	PostgreSQL, Redis, Elasticsearch, SQL Server, Firestore
Queues & Background Jobs	BullMQ, AWS SQS + DLQ, Sidekiq
Cloud & DevOps	AWS EC2 / S3 / ECS / ECR / Lambda / Cognito, Cloudflare Workers / R2 / D1, Docker, GitHub Actions, GitLab CI, Linux
Testing	RSpec, Vitest, pytest, Golden-vector regression tests
Payment Integrations	Stripe (Checkout, Subscriptions, Webhooks), PayPal Subscriptions, Fundiin, Payoo, ZaloPay, Recurring billing & proration, Webhook idempotency, Invoicing & consumption tax (適格請求書)
E-commerce Marketplaces	Shopee Open API v2, TikTok Shop API v2, Lazada Open Platform
Shipping & ERP	GHN, Ahamove, TikiNOW, NaviWorld ERP
Frontend	React, Next.js, Nuxt

EDUCATION

M.Sc. in Computer Science

2026 - present

Danang University of Science and Technology (DUT) - The University of Danang
Research track. Coursework in progress; thesis topic not yet decided.

B.Sc. in Information Technology

2015 - 2019

University of Science and Education - The University of Danang
Major in Information Technology.

RESEARCH

When Does Graph Memory Help Beyond Verbalized Relations? Isolating Representation from Structure in LLM-Agent Memory

2026 - present | Exploratory - no published results | Independent research - not committed as a thesis topic

Question. Graph memory is widely believed to help LLM agents remember better. But when graph memory is compared against flat memory, existing work changes TWO things at once: structure (the ability to traverse relations) and representation (information rewritten as typed relational sentences - unavoidable, because an LLM can only read text). So which of the two does the observed benefit belong to?

Method. Verbalize-control: three token-budget-matched conditions drawing from the same candidate edge pool and differing only in the selector - graph traversal, cosine truncated to the budget, and dump-everything. Plus a degree-preserving shuffled-edge control to separate “these particular edges” from “merely having a graph”. The design is pre-registered, evaluated on an external non-circular benchmark, and specifies the negative branch in advance (TOST equivalence testing).

Status, stated plainly: this is something I work on outside of client work, not a committed thesis topic. A 30-question feasibility probe has been run; the quality gate fired and returned “inconclusive” rather than a result. No publishable numbers.